Parsing Static Web Pages

Week 6 · Documents module · Textbook Chapter 6

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This week at a glance

We turn HTML — the language of web pages — into data you can analyze. Movies carry the examples all week: box office charts, Wikipedia film pages, and the Oscars.

Monday | Concepts & notebook intro

- Eight common shapes static pages hide their data in, and where each one sits on a single-site/multi-site, consistent/inconsistent design spectrum

Wednesday | Notebook lab

- Pilot each strategy in code: box office, Wikipedia, the Oscars, and Rotten Tomatoes

Friday | No class

- No class again on Friday, still submit PRs and issues for Sunday

Companion notebook

`ch-06-static-pages.ipynb`

Bring a laptop

Wednesday is hands-on — we debug live.

1. Monday · Concepts

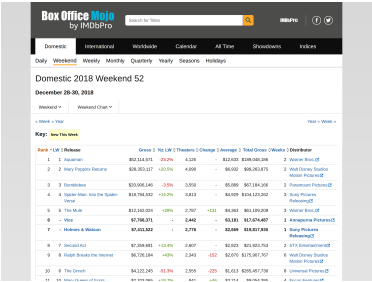
From HTML to data

Convert a structured format like HTML into another suited to analysis (DataFrame, CSV, or JSON)

While web pages might be structured, they rarely are in the structure you need to conduct an analysis

Static websites — we are going to focus on websites where the data arrives *in the HTML*, next weeks will examine more complicated cases of data arriving *after the HTML*

The core skill this week: eight common strategies for turning static websites into usable data



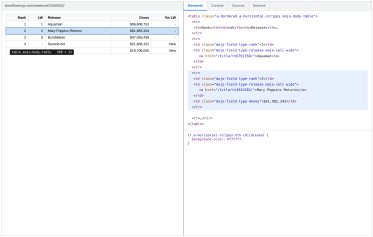
A page with the data already sitting in one table.

Inspect before you parse

Chapter 5 introduced the Inspector: right-click the data, choose **Inspect**, and read the DOM the browser actually built.

You already know your target is data in a page, but now you need to examine and understand the shape of the data to inform how to retrieve, clean, and analyze it

One habit from Chapter 5 carries forward unchanged: if an element is visible in the Inspector but missing from `requests.get(url).text`, JavaScript put it there after. Out-of-scope for this week but we will return next week.



The Elements panel on a real Box Office Mojo table.

Common research designs: a 2×2 spectrum

Before picking a parsing strategy, place your project on two axes: how many **sites** you're pulling from, and how **consistent** their design is page to page.

- **Single site, consistent design.** Fifty-two weekly Box Office Mojo pages — same template every time.
- **Single site, inconsistent design.** One oscars.org ceremony page but changes across years and nominees.
- **Multiple sites, consistent design.** English and French Wikipedia infoboxes — same platform, different specifics.
- **Multiple sites, inconsistent design.** Different movie sties will have unlrelated structures.

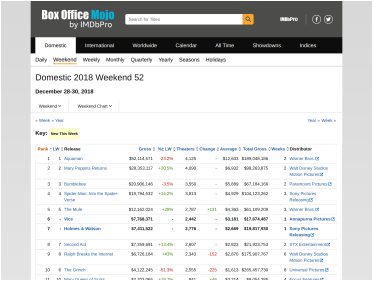
Strategy 1 — the clean single table

`boxofficemojo.com/weekly/2026W37/`

The best case: one table, one target, nothing else on the page competing for it.

Box Office Mojo — a single ranked table of that week's top movies. No navigation tables, no decoys, no ambiguity where the data lives.

A perfect use case for `pd.read_html()`.



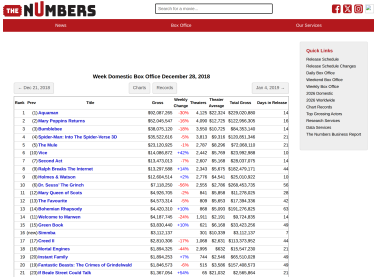
Box Office Mojo's weekly revenue numbers

Strategy 2 — a table hiding among decoys

`https://www.the-numbers.com/box-office-chart/weekend/2026/09/18`

The Numbers' weekly chart has the *same kind* of data as Box Office Mojo, but its HTML ships **two** `<table>` elements

You must inspect and count before you trust `find_all("table")[0]`.



Rank	Prev	Title	Gross	Weekly Change	Theaters	Theater Average	Total Gross	Days in Release
1	(1)	Aquaman	$92,087,288	-30%	4,129	$22,324	$229,020,808	14
2	(2)	Mary Poppins Returns	$52,045,547	-36%	4,208	$12,725	$122,066,306	16
3	(3)	Bumblebee	$38,075,120	-35%	3,958	$10,725	$64,563,148	14
4	(4)	Spider-Man: Into The Spider-Verse 3D	$30,522,614	-5%	3,813	$8,319	$103,951,348	21
5	(5)	The Mule	$23,120,505	-1%	2,707	$8,206	$72,966,119	21
6	(11)	Vice	$14,086,872	+42%	2,442	$5,768	$23,082,908	16
7	(7)	Second Act	$13,473,013	-7%	2,807	$5,208	$28,037,875	14
8	(9)	Ralph Breaks The Internet	$13,297,588	+14%	2,343	$5,675	$262,479,171	44
9	(8)	Holmes & Watson	$12,664,514	-2%	2,778	$4,561	$25,010,822	16
10	(10)	Dr. Seuss' The Grinch	$7,118,250	-50%	2,555	$2,786	$206,463,705	56
11	(12)	Mary Queen of Scots	$4,509,705	-2%	841	$5,359	$11,279,825	28
12	(13)	The Favourite	$4,573,314	-5%	808	$5,659	$17,284,339	42
13	(14)	Bohemian Rhapsody	$4,420,310	+10%	808	$5,693	$201,275,825	63
14	(21)	Welcome to Marwen	$4,187,745	-24%	1,611	$2,600	$9,724,835	14
15	(15)	Green Book	$3,930,440	+10%	621	$6,328	$33,423,258	48
16	(new)	Serenity	$3,112,137		301	$10,339	$3,112,137	7
17	(17)	Creed II	$2,833,306	-17%	1,868	$1,517	$113,573,952	44
18	(16)	Mortal Engines	$1,884,325	-44%	2,306	$817	$15,547,239	21
19	(20)	Instant Family	$1,884,253	+7%	744	$2,546	$65,510,828	49
20	(18)	Fantastic Beasts: The Crimes of Grindelwald	$1,846,573	-5%	515	$3,588	$257,480,573	49
21	(19)	If Beale Street Could Talk	$1,367,094	+44%	66	$21,820	$2,580,864	21

the-numbers.com — two tables, one is the data.

Strategy 3 — the table that needs cleanup

`https://en.wikipedia.org/wiki/List_of_highest-grossing_films`

Sometimes the table extracts cleanly but the values inside do not

Wikipedia's "List of highest-grossing films" is the classic case: `read_html()` succeeds on the first try, and the real work starts after.

Footnotes and superscript letters with complex hyperlinks inside sit alongside the values you care about. Numbers arrive as text.

Retrieval is easy, `pd.read_html()` or `soup.find_all()` both still work, but cleanup is gnarly

Rank	Peak	Title	Worldwide gross	Year	Ref
7	3	*Star Wars: The Force Awakens*	$2,068,223,624	2015	[# 12][# 13]
8	4	*Avengers: Infinity War*	$2,048,359,754	2018	[# 14][# 15]
9	6	*Spider-Man: No Way Home*	SM$1,922,598,800	2021	[# 16][# 17]
10	9	*Zootopia 2*	$1,866,647,950	2025	[# 18][# 19]
11	11	*The Odyssey* †	$1,721,314,740	2026	[# 20]
12	8	*Inside Out 2*	$1,698,863,816	2024	[# 21][# 22]
13	3	*Jurassic World*	$1,671,537,444	2015	[# 23][# 24]
14	7	*The Lion King*	$1,656,943,394	2019	[# 25][# 4]
15	3	*The Avengers*	$1,518,815,515	2012	[# 26][# 27]
16	4	*Furious 7*	$1,515,341,399	2015	[# 28][# 29]
17	11	*Top Gun: Maverick*	$1,503,260,455	2022	[# 30][# 31]
18	16	*Avatar: Fire and Ash*	$1,485,999,890	2025	[# 32][# 33]
19	10	*Frozen 2*	$1,450,026,933	2019	[# 34][# 35]
20	14	*Barbie*	$1,447,138,421	2023	[# 36][# 37]
21	5	*Avengers: Age of Ultron*	$1,402,809,540	2015	[# 38][# 29]

A table that parses fine and reads badly.

Strategy 4 — non-tabular structured data

`https://en.wikipedia.org/wiki/The_Godfather`

Not every page's real content lives inside a `<table>`

Wikipedia's film infoboxes are a labeled key–value block — "Directed by," "Produced by," "Starring" — one row per fact, in roughly the same shape on every film's page

Once you can pull one film's infobox apart, the same handful of lines pulls apart most others

The Godfather

Theatrical release poster

Directed by Francis Ford Coppola

en.wikipedia.org — *The Godfather*.

Strategy 5 — porting patterns to new sites

`https://fr.wikipedia.org/wiki/Le_Parrain_(film)`

The infobox idea from Strategy 4 isn't just reusable across English Wikipedia pages — it should work on other language editions

French Wikipedia runs the same software and lays out its film infobox the same way

But it is not the same page. The anchor that worked on the English article does not exist here at all; the reliable anchor turns out to be the table's `<caption>` instead.

The general strategy transfers but anchors/selectors may need to be re-verified

Titre original	*The Godfather*
Réalisation	Francis Ford Coppola
Scénario	Mario Puzo Francis Ford Coppola
Musique	Nino Rota
Acteurs principaux	Marlon Brando Al Pacino James Caan Richard S. Castellano Robert Duvall
Sociétés de production	Paramount Pictures Alfran Productions
Pays de production	États-Unis
Genre	gangsters
Durée	175 minutes
Sortie	1972

fr.wikipedia.org — *Le Parrain*, same idea.

Strategy 6 — cards that need to become rows

`https://www.oscars.org/oscars/ceremonies/2026`

The Academy Awards lists nominees and winners as a grid of individual cards with no table anywhere on the page.

Turning this into a DataFrame means treating each honoree as one row: find the repeating category container, then pull category name, winner-or-nominee status, person, and film out of each honoree inside it.

This is the shape behind most gallery and directory pages — and the one static-page pattern that shows up the most once you start looking for it.

oscars.org — 2026 ceremony, one card per honoree.

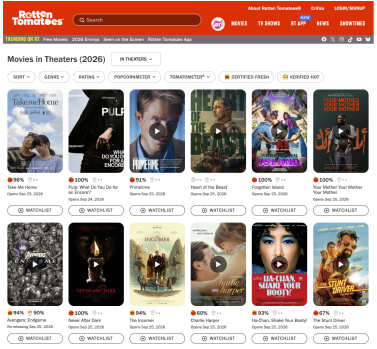
Strategy 7 — inconsistent cards, multiple sites

`https://www.rottentomatoes.com/browse/movies_in_theaters`

Rotten Tomatoes' "Movies in Theaters" grid contains a poster, a title, and a Tomatometer score, built from custom elements instead of plain `<div>`s.

A research question that needs *both* box office numbers and critical reception needs separate parsers for each site

The format and the content of the site can be customized via URL which *may* reload and change the HTML `rottentomatoes.com/browse/movies_in_theaters/sort:critic_highest`



rottentomatoes.com — a differently-shaped card.

Strategy 8 — when sites are not static

`https://www.imdb.com/title/tt0068646/`

All these strategies assumes the retrieved HTML already contains the data you want.

IMDb's own film pages break that assumption: a plain `requests.get()` against an IMDb title page returns HTTP **202** with **zero bytes** of body — not "missing the cast list," it does not exist in the HTML.

The real content is built by JavaScript after a browser loads the page, `requests` never runs JavaScript

We will return to this case of dynamically-loaded papges next week

What a browser renders. `requests` gets 0 bytes.

Which strategy?

Without writing any code come up with the strategy for scraping each of the websites below.

1. A film festival's "official selections" page
2. A movie theater's booking website
3. A list of recommended films from a streaming site
4. Or something else entirely

Use Inspection tab to evaluate where the data lives, which strategy is closest to what you'd need to turn it into tabular data, potential headaches, what the DataFrame would need to look like, what kinds of analyses you could run, why should anyone care about what you found

2. Wednesday · Notebook Lab

Strategy 1 — the one-line win

One table, one call. When the page cooperates, take the win:

```
import pandas as pd

url = "https://www.boxofficemojo.com/weekend/2018W52/"
tables = pd.read_html(url)
print(f"{len(tables)} table(s) found") # 1 -- no ambiguity here

weekly = tables[0]
print(weekly.head())
```

No `requests`, no `BeautifulSoup` — `read_html()` fetches the page itself. Save this reflex for pages that look like Strategy 1; the next one won't be this easy.

Strategy 2 — finding the real table among decoys

Retrieve, then find the right table (pages hide data tables among navigation or mobile-layout tables):

```
import requests
from bs4 import BeautifulSoup

url = "https://www.the-numbers.com/box-office-chart/weekly/2018/12/28"
headers = {"User-Agent": "WebDataScience/1.0 (you@colorado.edu)"}
soup = BeautifulSoup(requests.get(url, headers=headers).text, "html.parser")

tables = soup.find_all("table")
for i, table in enumerate(tables): # inspect each candidate
    print(i, len(table.find_all("tr")), "rows")
```

The one-line bug that shifts every column

Tempting shortcut:

```
# DON'T: silently drops empty cells
row.text.split("\n")
```

Correct — walk cell by cell so **empty cells stay aligned**:

```
cells = row.find_all(["td", "th"])
values = [c.get_text(strip=True)
          for c in cells]
```

A blank header cell (the “previous rank” column) is easy to lose.

Split-on-newline drops it. Every later value then shifts one column left, so **Movie** fills with distributor names.

The extra line of code buys correctness.

Strategy 3 — cleaning up after `read_html()`

```
import pandas as pd

url = "https://en.wikipedia.org/wiki/List_of_highest-grossing_films"
df = pd.read_html(url, header=0)[0] # verified by inspection: index 0
print(df.dtypes) # "Worldwide gross" parsed as text
```

Expect to spend as long **cleaning** as extracting:

```
# a stray footnote-reference letter can sit in front of the "$" --
# print a few raw values first so you know what you're stripping
df["gross"] = pd.to_numeric(
    df["Worldwide gross"].str.replace(r"[^0-9]", "", regex=True), errors="coerce")
df["footnote_count"] = df["Ref"].str.count(r"\[") # "[# 3][# 4]" -> 2
```

Strategy 4 — the infobox pattern

A table without a useful class name, but a very reliable shape — one `<th>`/`<td>` pair per row:

```
import requests
from bs4 import BeautifulSoup

url = "https://en.wikipedia.org/wiki/The_Godfather"
headers = {"User-Agent": "WebDataScience/1.0 (you@colorado.edu)"}
soup = BeautifulSoup(requests.get(url, headers=headers).text, "html.parser")

infobox = soup.find("table", class_="infobox")
facts = {}
for row in infobox.find_all("tr"):
    label, value = row.find("th"), row.find("td")
    if label and value:
        facts[label.get_text(strip=True)] = value.get_text(" ", strip=True)
print(facts["Directed by"]) # 'Francis Ford Coppola'
```

Strategy 5 — porting the parser to a new site

The class name `infobox` doesn't exist on this page at all. Re-inspect, and anchor on the `<caption>` instead — same loop as Strategy 4 after that:

```
url = "https://fr.wikipedia.org/wiki/Le_Parrain_(film)"
soup = BeautifulSoup(requests.get(url, headers=headers).text, "html.parser")

infobox = soup.select("table:has(caption)")[0] # the anchor that transfers
facts = {}
for row in infobox.find_all("tr"):
    label, value = row.find("th"), row.find("td")
    if label and value:
        facts[label.get_text(strip=True)] = value.get_text(" ", strip=True)
print(facts["Titre original"]) # 'The Godfather' -- same film, new labels
```

The loop is identical. Only the anchor and the labels changed.

Strategy 6 — cards to rows: the Oscars parser

oscars.org 403s an anonymous or browser-spoofed request — but not this honestly-identified one. Then navigate down through the real container hierarchy, category by honoree:

```
url = "https://www.oscars.org/oscars/ceremonies/2026"
soup = BeautifulSoup(requests.get(url, headers=headers).text, "html.parser")
area = soup.find("div", class_="field--name-field-award-categories")
categories = area.find_all("div", class_="paragraph--type--award-category")
rows = list()
for cat in categories:
    name = cat.find(class_="field--name-field-award-category-oscars").get_text(strip=True)
    for h in cat.find_all("div", class_="paragraph--type--award-honoree"):
        rows.append({
            "category": name,
            "type": h.find(class_="field--name-field-honoree-type").get_text(strip=True),
            "entity": h.find(class_="field--name-field-award-entities").get_text(strip=True),
        })
df = pd.DataFrame(rows) # 24 categories on the 2026 ceremony page
```

Strategy 7 — a different card, a different site

Rotten Tomatoes' tiles are custom elements, not `<div>`s — same idea, new anchors:

```
url = "https://www.rottentomatoes.com/browse/movies_in_theaters/"
soup = BeautifulSoup(requests.get(url, headers=headers).text, "html.parser")

tiles = soup.find_all("tile-dynamic", attrs={"data-qa": "tile"})
records = list()
for tile in tiles:
    title = tile.find("span", class_="p--small")
    score = tile.find("rt-text", class_="critics-score")
    records.append({
        "title": title.get_text(strip=True) if title else None,
        "tomatometer": score.get_text(strip=True) if score else None,
    })
# join records on title with the Oscars/box-office DataFrames -- one row per movie
```

Strategy 8 — watching `requests` fail, live

```python
import requests

url = "https://www.imdb.com/title/tt0068646/"
headers = {"User-Agent": "WebDataScience/1.0 (you@colorado.edu)"}
response = requests.get(url, headers=headers)
print(response.status_code, len(response.text))
# 202 0 -- zero bytes. Not a bad selector: there is nothing here to select.
```

This is not an error in your code

A 202 with an empty body means the server accepted the request and handed back a JavaScript shell instead of a page. `BeautifulSoup` has nothing to parse. This is the boundary this chapter's strategies don't cross — dynamic pages get their own week.

Make it reusable — and defensive

Abstract the working parser into a function; keep the `sleep` in the *caller*, not the function:

```
for year in range(2020, 2027):
    df = scrape_oscar_year(year, headers) # returns empty DataFrame on failure
    all_years.append(df)
    time.sleep(1) # politeness lives in the loop
```

Wrap fragile parsing so one bad page doesn't kill the run:

```
try:
    response.raise_for_status()
except requests.RequestException as e:
    print(f"Request failed: {e}"); return None
# AttributeError fires when .find() returns None and you call .text on it
# -- the single most common scraping error.
```

Error flavors

Some code here is *designed* to break (a guard demonstrating the `AttributeError` above) — that does not need a GitHub issue. Other code may be *genuinely* broken: confirm it, then search and file an issue.

Lab: work these, then show-and-tell

Work in pairs. Do these exercises from the notebook:

1. Get a Wikipedia table with `read_html`. Then make a visualization.
2. Write a parser for a page that has no table. Record your selectors.
3. Extend the parser to more than one page. Add a delay, error handling, and a function.
4. Parse a Wikipedia page that is not in English. Report the encoding problems that you correct.
5. Download each page one time into `cache/`. Then parse the files offline.
6. **5617**: examine the Terms of Service and the `robots.txt` file of two sites. Compare them with Fiesler et al. (2020)

Submitting

Start in class, finish at home, submit what you got working.

Graded on good-faith attempt to complete rather than getting everything working.

Show-and-Tell

Bring one of these to class: a bug you could not solve, data you find interesting, or a question for a research design.

Sites redesign without warning.
When a parser breaks, that's not a bug in your code —
it's the web working as designed.

Re-inspect, re-hypothesize, re-verify.

3. Friday · Textbook Revisions

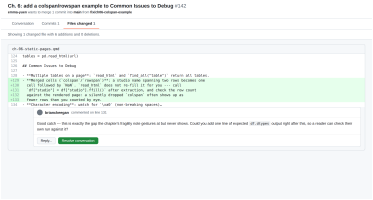
Improving Chapter 6 together

Today we make the book better. Each of you opens a **pull request** against `ch-06-static-pages.qmd` and reviews a classmate's.

The workflow (from Week 1):

1. Branch / fork the book repo
2. Edit the chapter; commit with a clear message
3. Open a PR describing *what* and *why*
4. Review two peers' PRs: kind, specific, actionable

Review comments still follow Week 3's claim / link / sharpen taxonomy — claim what you'll fix, link related issues, sharpen ones missing detail.



A revision menu for Chapter 6

High-value targets — pick one and scope it to a single PR:

- **Test the code against live sites.** The Numbers, Wikipedia, the Oscars, and Rotten Tomatoes all drift. Run the notebook, report what broke, and fix the selectors (this is the chapter's own "A Note on Fragility").
- **Close a gap in "Common Issues to Debug."** Add a worked `\xa0` / non-breaking-space example, or a `colspan/rowspan` case `read_html` mishandles.
- **Add a figure.** An annotated Inspector screenshot, or a diagram of the 2×2 research-design spectrum, would help readers place their own project.
- **Strengthen an exercise.** Add expected output, a rubric, or a starter cell so exercises are self-checking.
- **Extend "Further Reading."** A data-journalism scrape (ProPublica, The Markup) tied to the "Public Interest" callout.

A good revision, and a good review

A strong PR

- One focused change, clearly titled
- Explains the problem it fixes
- Builds (`quarto render`) without errors
- Matches the book's voice and notation
- Discloses any AI assistance

A strong review

- Runs / reads the change, not just skims
- Names specifics (line, claim, code)
- Separates "must fix" from "nice to have"
- Is generous — assume good faith
- Ends with a clear verdict

Next week: **Archived web pages** — scraping the past with the Wayback Machine.

Key takeaways

1. Scraping is a pipeline: **retrieve** → **inspect** → **anchor** → **extract** → **DataFrame**.
2. Place your project on the single/multi-site × consistent/inconsistent spectrum before picking a strategy.
3. Tables, decoys, infoboxes, and cards are different shapes of the same problem — finding a reliable anchor.
4. Abstract into functions, rate-limit in the loop, and code defensively.
5. Pages change: the durable skill is **inspect, hypothesize, verify** — not memorized class names.